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Coordinating Monetary and Macroprudential Policies

This paper studies noncooperative games between a monetary authority and a macroprudential regulator whose objectives are a subset of those in the social loss function. The analysis is based on a New Keynesian model with a financial sector and a financial friction à la Gertler and Karadi (2011). When the friction affects the financing of all factors of production equally, macroprudential policy is shown to be a powerful additional tool, fully eliminating inefficiencies, regardless of the source of the shock and no matter whether the central bank and the regulator cooperate. But when trade-offs are present and policy is discretionary, the institutional arrangements become crucial. While coordination leads to higher welfare than a setting in which each authority takes the decision rule of the other as given (namely, the Nash equilibrium), our analysis shows that a noncooperative setting in which the macroprudential authority acts as a leader within the period can be superior to cooperation. Finally, our conclusions are unaffected by whether the macroprudential instrument affects funding costs or acts as a liquidity requirement.

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“Let’s suppose that we have a house price bubble and the Financial Policy Committee increases capital requirements [...]. The Monetary Policy Committee then lowers the policy rate to push inflation back to target. Can we be confident that the lowering of the policy rate accompanying the widening in lending margins does not keep the house price boom going?”

SUSHIL B. WADHWANI

Former member of the Bank of England’s Monetary Policy Committee
Remarks at the London School of Economics, July 14, 2010

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RECENT DEVELOPMENTS IN FINANCIAL MARKETS and the world economy have motivated macroeconomists to think of new policy instruments that could help ensure financial stability. Central banks and financial regulators have begun their search for the appropriate macroprudential tool. But apart from finding the right instrument, policymakers are also interested in understanding how such instruments should be set in conjunction with monetary policy. Some commentators have suggested that separate committees for monetary and macroprudential policies could lead to coordination problems. For example, Wadhvani (2010) warned of the risk of “push-me, pull-you” behavior between policy committees as shown by the quotation in the epigraph. The Committee on International Economic and Policy Reform called for the Tinbergen separation principle to be retired (see Eichengreen et al. 2011). They stressed that rather than having one instrument devoted entirely to one objective, the macrostabilization exercise must be viewed as a joint optimization problem where monetary and regulatory policies are used in concert in pursuit of monetary and macroprudential objectives.

We provide a formal analysis of these coordination problems based on a micro-founded model. To do so, we incorporate an endogenous macroprudential tool into a version of the model of Gertler and Karadi (2011) (hereafter referred to as GK) in which there is a moral hazard problem between banks and depositors. Our framework encompasses the usual New Keynesian model with nominal rigidities and endogenous monetary policy. But, similar to Carlstrom, Fuerst, and Paustian (2010) (hereafter referred to as CFP) it also incorporates leverage constraints and endogenous spreads between deposit and lending rates, both arising from the moral hazard problem.

We derive a utility-based welfare criterion that characterizes the underlying distortions in this economy. In order to obtain a parsimonious expression for the loss function, we depart from GK by assuming that capital is fixed but there is variable capital utilization. The resulting quadratic loss function includes the typical terms in the variability of inflation and the output gap that are common in standard sticky price models. In addition, it also features the variation in credit conditions. Crucially, the credit distortion is given by the total cost of borrowing, namely, the sum of the short-term nominal interest rate and a spread. Movement in the nominal interest rate creates inefficiencies because it affects the borrowing costs of firms when financing part of their input costs as in Ravenna and Walsh (2005). At the same time, leverage of banks is restricted by depositors’ concern that their assets could be misused if leverage becomes too high. Consequently, bankers can charge a spread of loan rate over the deposit rate that ensures they have an incentive to use funds appropriately.

Macroprudential policy is modeled as a cyclical tax or subsidy on banks’ funding costs, while monetary policy sets the short-run nominal interest rate. We also consider an alternative macroprudential instrument that shares some similarities with liquidity restrictions, by assuming banks are required to invest a fraction of their deposits into other riskless assets instead of lending them. Our key focus is then on the interactions between the monetary and macroprudential instruments. Such interactions arise naturally in our framework. For instance, the impact of higher credit spreads on aggregate conditions and welfare could be directly offset by lowering the nominal interest. But

a particular path of the nominal interest rate that helps offset movements in the credit spread is, in general, not consistent with stabilizing the price level. Hence, there is a need for an additional policy instrument targeted directly at the credit friction.

In our analysis, we compare the case in which the monetary and macroprudential authority cooperate with the case of a noncooperative game between the institutions. Our focus is on equilibria under discretion. When studying the noncooperative setting, we assume that each authority has a loss function that depends on a subset of the goals found in the social welfare function. In our baseline setting, the loss of the central bank depends on the variability of inflation and the output gap while the loss of the macroprudential authority depends on the volatility of credit conditions and the output gap. We then allow the weights of these variables in the objective functions to vary and assess alternative policy mandates.

The policy mandates we consider have an attractive feature: they imply that the optimal noncooperative policy achieves the first best whenever this is possible under cooperation. The explanation for this feature is simple. In the noncooperative game, we assume that each authority cares about a subset of the welfare-relevant gap variables. Thus, if all welfare-relevant gaps are closed under cooperation, then no individual authority has any incentive to deviate from the rule under cooperation that pertains to its instrument given that this achieves a zero loss.

But if there are trade-offs under cooperation, then the cooperative and noncooperative equilibria differ and, in fact, the Nash equilibrium results in substantially lower social welfare than the coordinated equilibrium. Ironically, relative to the cooperative equilibrium, the targeting criterion in the Nash solution displays a more inflation-averse stance—which, by itself, tends to improve welfare in a discretionary equilibrium (as in Rogoff 1985). But the key shortcoming of the Nash equilibrium is that macroprudential policy fully stabilizes the total cost of borrowing for firms, rather than adjusting these costs to help achieve smaller fluctuations in the output gap. Regardless of whether the output gap is included in the loss function of the macroprudential authority, this authority perceives the output gap to be completely determined by the demand management of monetary policy. This is a form of coordination failure, and it creates nontrivial welfare losses.

The fact that the uncoordinated policy can lead the macroprudential authority to ignore output gap fluctuations can call into question the benefits of having a macroprudential instrument. We find that, in the face of markup shocks, two authorities that play a Nash game might arrive at a worst equilibrium than the one that a highly inflation-averse central bank with one instrument would achieve.

But the finding that introducing a macroprudential tool is harmful if used in noncooperative setting is by no means a general result. In fact, when firms have to borrow to pay for all factors of production, the macroprudential tool by itself can fully eliminate welfare losses following markup shocks—a job that a central bank setting interest rates would not be able to do.

In addition, the introduction of a macroprudential tool is always beneficial if the macroprudential authority acts as a “leader” when setting its instrument. A leadership setting can be seen as a reduced-form way of characterizing an environment in which

the two authorities may move at slightly different frequencies (e.g., quarterly for the leader, monthly for the follower). Our results suggest that leadership can improve social welfare. This is because the macroprudential authority recognizes that it can affect the evolution of the output gap via its influence on how the central bank sets interest rates. In order to avoid costly fluctuations in the output gap, it chooses to let firms' borrowing costs fluctuate. Leadership of the monetary authority, however, results in worse welfare outcomes than simultaneous choice of instruments (i.e., the Nash equilibrium).

Turning to the choice of macroprudential instruments, we find that an instrument that resembles a liquidity requirement acts in a similar fashion to a time-varying subsidy to deposits. Under a liquidity requirement, the bank is forced to invest a fraction of its deposits in riskless government bonds that carry a lower interest rate than loans to firms. We assume that the steady state effect of this instrument is fully offset and focus on the cyclical effects. Not surprisingly, increasing the liquidity requirement lowers the amount of loans extended to firms and banks become less leveraged. Overall, variations in the funding subsidy and the liquidity requirement act in a very similar way via their effects on loan supply. In this sense, our findings are robust to the different instrument specifications.

Finally, throughout our analysis, we revisit the result in Rogoff (1985) that outcomes under discretion can be improved by making the central bank more inflation-averse than is indicated by the social welfare. In our noncooperative game between the two institutions, we find that reducing the weight on the output gap in the central bank's loss function can substantially improve welfare, regardless of whether authorities play a Nash or a leadership game, and regardless of the policy instrument at hand. Our results reiterate the case for a conservative monetary policy mandate under discretion.

1. RELATED LITERATURE

There is a growing literature incorporating macroprudential policy in monetary models (see references in Galati and Moessner 2010, Beau, Clerc, and Mojon 2012). To the extent that this literature performs a welfare analysis, it typically assumes a coordinated setup with one institution setting all available policy tools. For example, Angeloni and Faia (2009) study a sticky price model with a fragile banking sector. Their analysis includes welfare-maximizing rules for capital ratios and the short-term interest rate. The authors find that capital ratios should be countercyclical and that monetary policy should respond to leverage and asset prices. Similarly, Darracq Paris, Soerensen, and Palenzuela (2010) analyze a model with a capital-constrained banking sector, but their focus is mainly positive. In the normative part of the paper, they consider a cooperative setting and determine optimal rules for monetary policy and capital ratios that maximize an *ad hoc* loss function.

Closer to our analysis is the paper by Collard et al. (2017), who study the jointly optimal plans for monetary and macroprudential policies in a model with limited

liabilities and deposit insurance. They outline the conditions under which prudential policies prevent inefficient risk taking, while monetary policy deals with the business cycle fluctuations. In that way, they provide what might be seen as a natural delegation of objectives between policy institutions.

While the recent literature has advanced in providing a rationale for the use of macroprudential tools, we aim at contributing to the debate by formalizing the interactions between such tools and monetary policy. In this respect, our approach is related to that of Dixit and Lambertini (2003), who study the interactions between monetary and fiscal policy under commitment and under discretion.

Our work is also related to that of Angelini, Neri, and Panetta (2012), who study the connections between monetary and macroprudential policies. A key contribution of theirs is an elaborate model of the banking sector. Nevertheless, having a more elaborate model comes at the cost of policy objectives that are not derived from the microfoundations of that model and thus are not clearly related to the underlying economic distortions.

Regarding the methodology, we use a linear-quadratic framework for optimal policy along the lines of Woodford (2003). This necessarily involves a financial constraint that is always binding. There is, nonetheless, a growing literature on optimal prudential policy that considers occasionally binding constraints (e.g., Jeanne and Korinek 2010, Bianchi 2011, and Otrok et al. 2012). This literature highlights the conditions under which it is beneficial from a welfare perspective to reduce borrowing in noncrisis times in order to reduce the likelihood and severity of a crisis (induced by a constraint that becomes binding). In this sense, such policies are truly prudential in nature. Our model abstracts from this distinction between “normal times” and “crisis times” and focuses on the business cycle implications of always-binding constraints. For the purpose of studying policy coordination, the linear-quadratic framework is more tractable than the nonlinear setup considered in these papers because the linear-quadratic approach allows for a simple characterization of welfare in terms of stabilization objectives that is amenable to delegation between different authorities.

2. MODEL

The core of the model is a new Keynesian sticky price cashless economy without capital accumulation as popularized by Woodford (2003). In addition, we assume a costly enforcement problem as in Gertler and Karadi (2011). In particular, the possibility of banks diverting funds from depositors implies that banks are constrained in the amount they can lend to firms. This financial friction motivates the use of macroprudential instruments. In what follows, we consider two regulatory tools that affect the total amount of lending by banks. Our benchmark model assumes that there is a tax/subsidy to banks’ funding costs. We then consider the case in which the policy authority imposes that banks have to hold a certain amount of liquid assets in their balance sheets.

2.1 Households

As in GK, within each household there are workers, who supply labor, and bankers, who manage a financial intermediary. Earnings of both workers and bankers are transferred back to the household and there is perfect consumption insurance within the family. This allows the model to feature financial friction while retaining its representative agent nature. At any point in time a fraction f ($1 - f$) of household members are workers (bankers) and individuals switch between the occupations. Moreover, a banker this period remains a banker next period with probability γ . The proportion of bankers/workers in each period is constant and, while bankers transfer their earnings to the household upon exit, workers provide startup funds to new bankers.

The typical household consumes the final good c_t and supplies labor L_t and capital utilization services u_t to the entrepreneurs. For simplicity, capital is assumed to be fixed with variable—and costly—utilization given by u_t . Standard models assume that more intense utilization causes heavier depreciation. But this representation would not be feasible in a model without capital accumulation. So, as an alternative, we assume that heavier utilization causes a utility cost much like working longer hours. Else, one could interpret u_t as a second type of labor as in CFP. The factor prices of labor and utilization services are w_t and r_t , respectively. Preferences are given by

$$U(c_t, L_t, u_t) = \frac{c_t^{1-\sigma}}{1-\sigma} - \frac{L_t^{1+\theta}}{1+\theta} - \frac{u_t^{1+\theta}}{1+\theta}.$$

At period t , households make deposits A_t , decide their consumption expenditure $P_t c_t$ at the financial intermediary, receive their return for each input $P_t(w_t L_t + r_t u_t)$, and receive interest on their deposits ($R_{t-1} A_{t-1}$). As in GK, deposits of a given household are held in intermediaries managed by bankers belonging to a different household. Households' budget constraint, written in real terms, is

$$c_t + A_t = \Omega_w w_t L_t + \Omega_r r_t u_t + \frac{R_{t-1}}{\pi_t} A_{t-1} + T_t + \Pi_t,$$

where the inflation rate is defined as $\pi_t \equiv \frac{P_t}{P_{t-1}}$ and Π_t is the payout to households from exiting bankers net of the startup funds transferred to new bankers. Ω_w and Ω_r are subsidies to factor payments financed by the lump sum taxes, T_t . Motivation for these subsidies is provided in a later section when we discuss the welfare-based loss function faced by the policymakers.

The first-order conditions for labor supply, utilization services, and bond holdings are

$$\frac{L_t^\theta}{c_t^{-\sigma}} = \Omega_w w_t, \quad (1)$$

$$\frac{u_t^\theta}{c_t^{-\sigma}} = \Omega_r r_t, \quad (2)$$

$$c_t^{-\sigma} = \beta E_t \left(c_{t+1}^{-\sigma} \frac{R_t}{\pi_{t+1}} \right). \quad (3)$$

2.2 Intermediate Good Producers

Intermediate firms hire both capital and labor from households and use these inputs in a constant returns to scale production function to produce an intermediate good. But in order to pay the wage bill associated with the input L_t , entrepreneurs have to borrow $b_t = w_t L_t$ from intermediaries at the rate R_t^B , as in Ravenna and Walsh (2005), in advance of selling their goods. Firms' profits are thus given by

$$\text{profits}_t = p_t x_t - R_t^B w_t L_t - r_t u_t,$$

where p_t denotes the relative price of the intermediate and $x_t = L_t^\alpha u_t^{1-\alpha}$ denotes production of the intermediate good. The parameter α governs the importance of financial frictions for production in this model. If $\alpha = 0$, the only input factor is the one for which there is no need to borrow in advance. Consequently, the model reduces to the simple New Keynesian sticky price model. The first-order conditions for the choice of input factors are

$$\alpha p_t x_t = (1 + \phi_t) R_t w_t L_t, \quad (4)$$

$$(1 - \alpha) p_t x_t = r_t u_t, \quad (5)$$

where ϕ_t is a measure of the spread between the borrowing rate R_t^B and the deposit rate. In particular, we define $\phi_{t-1} \equiv (R_t^B - R_{t-1})/R_{t-1}$. Intermediate firms, owned by households, make no profits.

2.3 Final Good Production and Market Clearing

Monopolistically competitive firms indexed by j produce final goods $y_{t,j}$ that are aggregated to an output bundle according to a constant elasticity of substitution aggregator

$$y_t = \left[\int_0^1 y_{t,j}^{\frac{\epsilon_t-1}{\epsilon_t}} dj \right]^{\frac{\epsilon_t}{\epsilon_t-1}},$$

with $\epsilon_t > 1$ assumed to be a time-varying markup parameter. The final goods firms purchase the intermediate good from entrepreneurs at relative price p_t . The production function is given by $y_{j,t} = a_t x_{t,j}$, where productivity a_t follows an exogenous first-order autoregressive process in logs. Since the production function is linear, real marginal cost is given by

$$z_t = \frac{p_t}{a_t}.$$

As in Rotemberg (1982), there are quadratic costs of price adjustment that enter the profit function of firm j as $\frac{\phi}{2}[(p_{t,j} - p_{t-1,j})/p_{t-1,j}]^2 y_t$, with $\phi > 0$. These costs disappear from the linearized version of the social resource constraint if the steady-state inflation rate is zero. In a symmetric equilibrium, the Rotemberg price setting problem gives rise to a Phillips curve relationship:

$$0 = (1 - \varepsilon_t) + \varepsilon_t z_t - \varphi(\pi_t - 1)\pi_t - \beta E_t \left[\frac{c_{t+1}^{-\sigma}}{c_t^{-\sigma}} \varphi(\pi_{t+1} - 1)\pi_{t+1} \frac{y_{t+1}}{y_t} \right]. \quad (6)$$

Monopolistic competition implies that firms earn profits in equilibrium. These profits are paid out as dividends to shareholders of the sticky-price firms. These dividends are given by

$$d_t = y_t(1 - z_t) - \frac{\varphi}{2}(\pi_t - 1)^2 y_t. \quad (7)$$

In a symmetric equilibrium, where $y_{jt} = y_t \forall j$, aggregate output is given by

$$y_t = a_t L^\alpha u_t^{1-\alpha}.$$

The market clearing condition for the final good requires total output equals consumption and price adjustment costs.

$$y_t = c_t + \frac{\varphi}{2}(\pi_t - 1)^2 y_t.$$

2.4 Financial Intermediaries

In each period, financial intermediaries—or banks—lend to intermediate good producers at a nominal interest rate R_t^B and remunerate households' deposits at a rate R_t , subject to a subsidy S_t . A net worth of a given banker j is given by

$$N_{jt} = R_{t-1}^B B_{jt-1} - R_{t-1}(1 - S_{t-1})A_{jt-1}, \quad (8)$$

where $B_t = P_t b_t$ is the nominal value of loans. Given banks' balance sheet identity $B_t = N_t + A_t$, we can write

$$N_{jt} = (\phi_{t-1} + S_{t-1})R_{t-1}B_{jt-1} + (1 - S_{t-1})R_{t-1}N_{jt-1}. \quad (9)$$

As long as the bank can charge a loan rate that is higher than the deposit rate, the bank will want to accumulate assets until exiting the industry. Consequently, the bankers' objective is to maximize their terminal wealth, given by

$$V_{jt} \equiv \max E_t \sum_{i=0}^{\infty} (1 - \gamma) \gamma^i \beta^i \Lambda_{t,t+1+i} N_{jt+1+i}. \quad (10)$$

We can write this recursively, as

$$V_{jt} = \max \left\{ E_t(1 - \gamma)\Lambda_{t,t+1}N_{jt+1} + E_t\gamma\beta\Lambda_{t,t+1}V_{jt+1} \right\}, \quad (11)$$

where $\Lambda_{t,t+1} = \frac{c_{t+1}^{-\sigma}}{c_t^{-\sigma}\pi_{t+1}}$ is the discount factor of bankers at time t that applies to earnings at time $t + 1$. As in GK, we introduce the following costly enforcement problem: In each period, bankers can divert a fraction of assets and transfer it back to the household to which they belong. In particular, we assume that the bank can run away with a fraction λ_t of the loans and that there may be shocks to such fractions. This shock captures exogenous variation in the strength of the moral hazard problems. Bankers will choose not to divert the funds if the value of continuing to operate outweighs the benefits of diverting the assets. This generates a market-based leverage constraint where depositors will only supply funds if the following incentive compatibility constraint is satisfied:

$$V_{jt} \geq \lambda_t B_{jt}. \quad (12)$$

Following GK, we can express V_{jt} as follows

$$V_{jt} = v_{b,t} B_{jt} + v_{n,t} N_{jt}, \quad (13)$$

where

$$v_{b,t} = (1 - \gamma)(\phi_t + S_t) + E_t\gamma\beta\Lambda_{t,t+1}\pi_{b,t+1}v_{b,t+1}, \quad (14)$$

$$v_{n,t} = (1 - \gamma)(1 - S_t) + E_t\gamma\beta\Lambda_{t,t+1}\pi_{n,t+1}v_{n,t+1}, \quad (15)$$

where $\pi_{b,t+1} \equiv \frac{B_{jt+1}}{B_{jt}}$, and $\pi_{n,t+1} \equiv \frac{N_{jt+1}}{N_{jt}}$.

So, assuming that this constraint is always binding, combining equations (12) and (13) and defining bankers' leverage as $\delta_t \equiv \frac{B_{jt}}{N_{jt}}$, we have

$$\delta_t = \frac{v_{n,t}}{\lambda_t - v_{b,t}}, \quad (16)$$

so the shock λ_t affects the ability of banks to leverage themselves. Substituting for this expression in equation (9), we have

$$N_{jt} = [(\phi_{t-1} + S_{t-1})\delta_{t-1} + 1 - S_{t-1}]R_{t-1}N_{jt-1}, \quad (17)$$

$$\pi_{n,t} \equiv [(\phi_{t-1} + S_{t-1})\delta_{t-1} + 1 - S_{t-1}]R_{t-1}, \quad (18)$$

$$\pi_{b,t} \equiv \frac{\delta_t}{\delta_{t-1}}\pi_{n,t}. \quad (19)$$

We can now derive the evolution of total net worth $N_t = N_t^e + N_t^n$, where the superscript e denotes existing bankers and n denotes new bankers.¹ Given the probability γ of bankers remaining in the industry until next period, from equation (17) we can express existing bankers' net worth as

$$N_t^e = \gamma[(\phi_{t-1} + S_{t-1})\delta_{t-1} + 1 - S_{t-1}]R_{t-1}N_{t-1}. \quad (20)$$

Moreover, given the assumption that households provide startup funds to new bankers, as in GK, we can write their net worth as

$$N_t^n = \omega B_{t-1}.$$

Thus, allowing for a net worth shock N_t^s , we can express the evolution of aggregate net worth as

$$N_t = \gamma[\phi_{t-1}\delta_{t-1} + 1 - S_{t-1}(1 - \delta_{t-1})]R_{t-1}N_{t-1} + \omega B_{t-1} + N_t^s. \quad (21)$$

2.5 Policy

The model is closed by a description of policy behavior. In subsequent sections, we will consider different assumptions for how policymakers set their monetary and macroprudential instruments, R_t and S_t . The main focus of our analysis will be on the case of optimal policy. But in doing so, we will analyze the case in which the monetary authority and the macroprudential authority work in a cooperative as well as a noncooperative fashion. We will also consider the case in which policymakers can commit to a policy plan and the case in which they act under discretion.

2.6 Summary of Equilibrium Conditions

We assume that $\omega = 0$. Theory only requires some positive endowment for newly starting bankers, but any small endowment would suffice. Hence ω can be arbitrarily small. But then the model dynamics are practically unaffected and hence we set $\omega = 0$ for convenience of obtaining a simple log-linear system. In this case, the system can be reduced to:

$$\hat{\pi}_t = \kappa [(\sigma + \theta)\hat{y}_t^g + \alpha(\hat{R}_t + b\hat{\phi}_t) + \hat{\varepsilon}_t] + \beta E_t \hat{\pi}_{t+1}, \quad (22)$$

$$\hat{R}_t = \sigma E_t \Delta \hat{y}_{t+1}^g + \frac{\theta + 1}{\sigma + \theta} \sigma E_t \Delta \hat{a}_{t+1} + E_t \hat{\pi}_{t+1}, \quad (23)$$

$$\hat{n}_t = \hat{n}_{t-1} + \hat{R}_{t-1} - \hat{\pi}_t + \frac{1}{\phi\delta + 1} [\phi\delta\hat{\delta}_{t-1} + \delta\hat{\phi}_{t-1} + (\delta - 1)\hat{S}_{t-1}] + \hat{n}_t^s, \quad (24)$$

1. From equations (14), (15), (18), and (19), we can see that leverage δ_t does not depend on bank-specific factors, hence it does not have a subscript j .

$$\widehat{\delta}_t + \widehat{n}_t = (1 + \sigma + \theta)\widehat{y}_t^g + \frac{\theta + 1}{\sigma + \theta}\widehat{a}_t - (1 - \alpha)(\widehat{R}_t + b\widehat{\phi}_t), \quad (25)$$

$$\widehat{\delta}_t + \widehat{\lambda}_t = \delta\widehat{\phi}_t + (\delta - 1)\widehat{S}_t + \beta E_t [(\phi\delta + 1)\widehat{\delta}_{t+1} + \widehat{\lambda}_{t+1}], \quad (26)$$

where variables with a hat are in log-deviations from steady state, with the exception of $\widehat{\phi}_t$, $\widehat{S}_t$, and $\widehat{R}_t$, which are defined as $\widehat{\phi}_t \equiv \phi_t - \phi$, $\widehat{S}_t \equiv S_t - S$. Moreover, the output gap is defined as $\widehat{y}_t^g \equiv \widehat{y}_t - \frac{1+\theta}{\sigma+\theta}\widehat{a}_t$, $\widehat{n}_t$ denotes the real net worth, and $b \equiv \frac{1}{1+\phi}$.

The Phillips curve equation (22) illustrates how both the cost channel and the credit channel introduce a form of endogenous markup into the model. That is, cost-push fluctuations arise from movements in the exogenous shock ϵ_t^π as well as fluctuations in the effective interest rate, given by $\widehat{R}_t + b\widehat{\phi}_t$ and hereafter denoted by $\widehat{f}_t$. Equation (23) in the system above represents the economy's Euler, or IS, equation. Equation (24) determines the evolution of bankers' net worth where $\widehat{n}_t^s$ denotes an exogenous shock to banks net worth. Equation (25) is the firm's labor demand equation linked with the banking sector variables by the requirement that wages must be financed by loans. Finally, (26) is derived from the incentive compatibility constraint, which stipulates that banks can become more leveraged today if they can earn a larger loan–deposit spread $\widehat{\phi}_t$ or a larger deposit subsidy $\widehat{S}_t$ in the current and future periods.

The model is closed by a description of policy behavior. In subsequent sections, we will consider different assumptions for how policymakers set their monetary and macroprudential instruments, $\widehat{R}_t$ and $\widehat{S}_t$. The main focus of our analysis will be on the case of optimal policy. As a benchmark—showing the best available policy—we consider the case in which the monetary authority and the macroprudential authority can commit to a policy plan and act in a cooperative fashion. Under discretion, we consider the case in which they work in a cooperative as well as a noncooperative fashion.

3. SOCIAL WELFARE LOSSES

In our model, the presence of nominal rigidities, credit constraints, and exogenous cost-push shocks generates inefficient economic dynamics. We highlight the welfare costs associated with these distortions by deriving a quadratic microfounded loss function. Such a quadratic welfare criterion is useful not only because it highlights the sources of economic distortions but also because it can be evaluated accurately up to second order by considering linearized equilibrium conditions.²

If we assume, as suggested above, that the time-invariant subsidies Ω_w and Ω_r are set such that the welfare-relevant variables are efficient in the steady state,³ we can

2. See Woodford (2003) for details.

3. We assume that these subsidies are only available to correct average distortions but cannot be used as state-contingent stabilization tools. This assumption is based on practical considerations, namely, that such fiscal policy instruments are currently not likely to be available for stabilizing the distortions created by credit frictions and inflation. However, if such state-contingent tools were available to the fiscal authority,

derive the following quadratic welfare criterion:

$$\mathcal{L}_t = E_t \sum_{i=0}^{\infty} \beta^i \left[\frac{1}{2} \hat{\pi}_{t+i}^2 + \chi_y (\hat{y}_{t+i}^g)^2 + \chi_f (\hat{f}_{t+i})^2 \right], \quad (27)$$

where $\chi_y \equiv \frac{(\sigma+\theta)}{\varphi}$, $\chi_f \equiv \frac{\alpha(1-\alpha)}{\varphi(1+\theta)}$, and $\hat{f}_t \equiv \hat{R}_t + b\hat{\phi}_t$. The first term in equation (27) highlights the presence of relative price distortions. The second represents deviations of output from its efficient level, which in our model are driven by markup shocks and the presence of a credit friction. Moreover, when $\alpha \neq 1$ fluctuations in the credit spread also generate misallocation of resources between the different factors of production—since only a fraction of those is subject to credit constraints. These are represented by the last term in (27). If we set $\alpha = 1$, the expression for the loss function looks identical to the one in Ravenna and Walsh (2005), in which welfare is affected by a cost channel—namely, the fact that firms have to borrow in advance—and nominal rigidities. But even with $\alpha = 1$, fluctuations in output away from its efficient level (i.e., those represented by the second term in the loss function) in our model will differ to the ones in Ravenna and Walsh (2005) because of the presence of a credit friction. Note that although our framework explicitly models financial intermediation, the resulting quadratic loss function is similar to the one obtained by CFP. The sole difference comes from the presence of the cost channel, which introduces distortions coming from fluctuations in the risk-free interest rate, $\hat{R}_t$.

4. OPTIMAL POLICY

In what follows we study different ways in which a monetary and a macroprudential authority set their policy instruments. We start by analyzing the environment in which the monetary and macroprudential authorities cooperate when setting these instruments. That is, they jointly choose the nominal interest rate, $\hat{R}_t$, and the macroprudential subsidy, $\hat{S}_t$, as to minimize the social loss function (27). We then move to our analysis of the noncooperative equilibrium—that is, the case in which each policy authority sets its instrument optimally to minimize its individual loss function taking as given a reaction function for the instrument of the other authority. A discussion of how the loss functions are delegated is presented in Section 4.3. Throughout our analysis, we focus on the case in which the policymakers act under discretion. We nevertheless present the solution of the cooperative problem with commitment as to represent the best outcome that can be achieved with the policy tools available. As sensitivity analysis in Section 5, we briefly discuss the noncooperative equilibrium under commitment as well as the use of a different macroprudential tool.⁴

results similar to those obtained in Correia, Nicolini, and Teles (2008) would go through: namely, that certain distortions do not affect the set of implementable allocations if the right state-contingent instruments exist. A similar point has recently been made in the context of the zero lower bound by Correia et al. (2011).

4. A detailed description of the optimization problem in the different settings is described in our Technical Appendix available upon request.

TABLE 1
CALIBRATION

Parameter	Interpretation	Value
β	Discount factor	0.99
σ	Risk aversion	1
α	Labor share	0.5
χ	Inverse labor supply elasticity	0.47
ϵ	Elasticity of bundler	10
δ	Steady-state leverage	9
ϕ	Steady-state credit spread	2%
φ	Price adjustment costs	211
ρ_a	AR(1) TFP process	0.95
ρ_m	AR(1) markup process	0.95
ρ_n	AR(1) net worth process	0
$stdv(u_n)$	stdv of net worth shocks	0.043
$stdv(u_a)$	stdv of technology shocks	0.005
$stdv(u_m)$	stdv of markup shocks	0.005

4.1 Calibration and Numerical Algorithm

Although most of our results are shown analytically, in order to illustrate the implication of the different policy settings for welfare and economic dynamics, we also solve the model numerically. Our computational framework builds upon the algorithm of Dennis (2007) to solve for the Nash equilibrium for the game between the two policy authorities under discretion. We extend this algorithm to allow for a within period leadership setting.⁵

The parameter values chosen for our benchmark calibration are shown in Table 1. We assume log utility and, thus, a unitary elasticity of intertemporal substitution. The discount factor β is set at 0.99. The elasticity of substitution between the differentiated goods ϵ is set to 10, which results in a monopolistic markup of around 11%. The Rotemberg price adjustment cost φ is set to 173.08, implying that prices stay fixed for an average of five quarters in an equivalent Calvo setting. The share of the input factor into production that require borrowing is set to 0.5.

Following Woodford (2003), we assume a unitary elasticity of labor supply. We assume that the total factor productivity, markup, and incentive constraint shocks follow an AR(1) process with autoregressive coefficient 0.95, whereas net worth shocks are assumed to be white noise. Turning to the parameters governing the financial intermediary sector, we assume that the average spread of loan rate over deposit rate, ϕ , is 2%, while the average leverage of intermediaries, δ , is 9—consistent with a capital ratio of 11%. We calibrate the standard deviation of shocks by simulating the model under the assumption of no active macroprudential tool and a standard Taylor rule, with the coefficient governing the interest rate response to inflation and the output gap set to 1.5 and 0.125, respectively. We then choose volatility of shocks to match the variance of the inflation, output gap, and credit spreads. The variance of output and credit spreads is taken from Kollmann (2013), who computes

5. A Technical Appendix describing the numerical algorithm is available upon request.

the credit spread as the difference between commercial/industrial loan rates and the federal funds rate (as available from the Federal Reserve Board's Survey of Terms of Business Lending). The variance of inflation is calculated from quarterly U.S. nonseasonally adjusted CPI from 1990 to 2010.

4.2 Cooperation

Commitment. We first consider the case in which the monetary and the macroprudential authorities set policy under commitment—that is, the authorities have the ability to deliver on past promises. In particular, policymakers commit to a path of $\widehat{R}_t$ and $\widehat{S}_t$ as to maximize the discounted sum of period utilities given by (27), subject to the equilibrium conditions (22)–(26). The first-order conditions of the policy problem imply that

$$\widehat{\pi}_t = -\frac{\chi_y}{\kappa(\sigma + \theta)} \Delta \widehat{y}_t^g, \quad (28)$$

$$\widehat{f}_t = \frac{\alpha \chi_y}{\chi_f(\sigma + \theta)} \widehat{y}_t^g. \quad (29)$$

We can think of the joint policymaking body choosing $\{\pi_t, y_t^g, f_t\}$ directly, and the expressions can be thought of as optimal “targeting rules.” Equation (28) highlights the optimal trade-off between output-gap growth and inflation. Alternatively, as discussed in Gali (2008), equation (28) implies that there is trade-off between the price level and the output gap. Equation (29) determines the optimal trade-off between the effective interest rates and the output gap. As previously mentioned, the equilibrium under commitment and cooperation is a useful benchmark since it represents the best outcome that can be achieved given the constraints of the model and the two degrees of freedom given by the policy tools.

Discretion. We now consider the case in which the monetary and the macroprudential authorities set policy under discretion. In a discretionary equilibrium, the current policymaker maximizes its objective function, taken as given the decision rule of future policymakers and of the private sector. Since the policy problem is time-invariant, the policy function of current and future policymakers coincide in equilibrium. We assume that the policymaker moves before the private sector in any given period.

We restrict our attention to Markov policies, that is, policies that are linear functions of a minimal set of payoff relevant state variables as in the works of Kydland and Prescott (1977), Cohen and Michel (1988), and Currie and Levine (1993).⁶

If we define the following variable

$$x_{t-1} \equiv \widehat{n}_{t-1}^r + \widehat{R}_{t-1} + \frac{1}{\phi\delta + 1} [\delta\phi\widehat{\delta}_{t-1} + \delta\widehat{\phi}_{t-1} + (\delta - 1)\widehat{S}_{t-1}], \quad (30)$$

6. This rules out trigger strategies as well as policies that depend on nonpayoff-relevant predetermined variables as, for instance, in Hahn (2014). The concept of Markov equilibrium is attractive because it is parsimonious.

we can write the system of equilibrium conditions (22) - (26) as having a sole endogenous state variable x_t . The current policymaker can thus influence future outcomes via its effect on this state. Letting $z_t \equiv \{\widehat{a}_t, \widehat{e}_t, \widehat{n}_t^s, \widehat{\lambda}_t\}$ denote the vector of exogenous shocks, and V_{t+1} the quadratic value function, the problem of the policymaker at time t is

$$\min_{R_t, S_t} V_t(x_{t-1}, z_t) = \widehat{\pi}_{t+i}^2 + \chi_y (\widehat{y}_{t+i}^g)^2 + \chi_f \widehat{f}_{t+i}^2 + \beta E_t V_{t+1}(x_t, z_{t+1})$$

subject to the constraints given by (22)–(26) where future variables are expressed as linear functions of the states with coefficients to be determined.

We guess and then verify that there exists a cooperative equilibrium, where $\{\widehat{\pi}_t, \widehat{y}_t^g, \widehat{f}_t\}$ —and hence V_t —do not depend on the endogenous state variable. Under that conjecture, one can show that the Lagrange multipliers on equations (23)–(26) are zero. Thus, the problem under cooperation boils down to solving the following Lagrangian problem:

$$\mathcal{L} = \widehat{\pi}_t^2 + \chi_y (\widehat{y}_t^g)^2 + \chi_f \widehat{f}_t^2 + \gamma_t [\widehat{\pi}_t - \kappa(\sigma + \theta)\widehat{y}_t^g + \kappa\alpha\widehat{f}_t]. \quad (31)$$

The first-order conditions can be combined to yield the following two targeting criteria:

$$\widehat{\pi}_t = -\frac{\chi_y}{\kappa(\sigma + \theta)} \widehat{y}_t^g \quad (32)$$

$$\widehat{f}_t = \frac{\alpha\chi_y}{\chi_f(\sigma + \theta)} \widehat{y}_t^g. \quad (33)$$

While in the commitment solution there is a trade-off between the output gap and the price level, in the optimal discretionary policy there is a trade-off between inflation and the output gap. For example, following a markup shock, the discretionary solution takes the form $\widehat{\pi}_t = \theta_\epsilon^\pi \widehat{e}_t$ and $\widehat{y}_t^g = -\theta_\epsilon^y \widehat{e}_t$ while the commitment solution implies $\widehat{y}_t^g = \rho_y \widehat{y}_{t-1}^g - \widehat{\theta}_\epsilon^y \epsilon_t^\pi$. Implicit in the latter is the fact that output is kept away from target even after inflation has stabilized. This generates movements in inflation expectations that reduce how much inflation fluctuates for any given deviation of output from its efficient level (when compared with the discretionary equilibrium). The trade-off between the output gap and effective interest rates is the same under discretion and commitment but, clearly, the trade-off between inflation and the effective interest rates differs.

As shown in Table 2, when the authorities cooperate the welfare cost of lack of commitment is about 0.002%, measured in terms of steady-state consumption equivalent. That is, in order for agents to be indifferent between the discretionary and the commitment outcome, they would have to receive 0.002% of their (steady-state) consumption every period, or equivalently, they would need to receive a one-time transfer of 0.2% of their steady-state consumption today. This number is in the same order of

TABLE 2
WELFARE LOSSES UNDER BENCHMARK ASSIGNMENTS OF MANDATES AND OPTIMIZED WEIGHTS IN THOSE MANDATES

	Welfare losses benchmark	Optimal weights	Welfare losses optimal
Commitment coop.	0.0028	<i>N.A.</i>	<i>N.A.</i>
Discretion			
No macroprudential	0.0208	$\chi_y^*=0, \chi_f^*=0.52$	0.0056
Cooperation	0.0044	$\chi_y^*=0.06, \chi_f^*=0.06$	0.0029
Nash	0.0075	$\chi_y^{cbs}=0.11, \chi_y^{mps}=indet$	0.0059
Monetary leader	0.0113	$\chi_y^{cbs}=0.06, \chi_y^{mps}=indet$	0.0059
Macroprudential leader	0.0038	$\chi_y^{cbs}=0.12, \chi_y^{cbs}=1.1$	0.0029

NOTES: The monetary policy objective is social welfare in the absence of a macroprudential tool χ_y^{cbs} denotes the optimal weight on the output gap relative to inflation in the objective function of the central bank, while χ_y^{mps} denotes the optimal weight on the output gap relative to the financial distortion. The entry indent indicates that welfare is independent of this relative weight.

magnitude as the one found in Lucas (1987) when measuring the costs of business cycles. These welfare losses associated with discretion are contingent on the fact that the objective of the policy authorities is to minimize the social loss function (27). But, in a second-best world in which policymakers cannot commit, it is not clear that assigning such a policy mandate would lead to the best outcome. In fact, the monetary literature argue that assigning the monetary authority, a larger concern for inflation stabilization than what is indicated by the true social welfare function can improve welfare if the central bank acts under discretion. This case for a “conservative central banker” was first highlighted by Rogoff (1985). The intuition is that this large concern for inflation stabilization mitigates the discretionary bias that arises from the inability of the central bank to refrain from exploiting predetermined inflation expectations in order to reduce output volatility. But when the concern for inflation stabilization is large relative to the concern for output volatility, there is little desire by the discretionary central bank to reduce output gap volatility in the first place. In this sense, narrow mandates—or a bigger focus on price stability—can partially substitute for the inability to commit. The results in Table 2 demonstrate that a similar argument holds in our setting. In particular, the last column of the table shows that, if one could choose the weights χ_y and χ_f optimally under discretion, these weights would be small and identical ($\chi_y^* = \chi_f^* = 0.06$). The small weights reduce the inflationary bias due to discretion, as discussed above. In addition, the optimal trade-off between the effective interest rate and the output gap should be dictated by their cost push implication, which is given by their relative weight in the Phillips curve ($\alpha/(\sigma + \theta)$). From equation (33), one can see that this is achieved by assigning equal weights to both variables. The discretionary outcome with such optimal weights mimics closely the one under commitment, with the costs associated with discretion decreasing to 0.0001% of steady-state consumption.

Achieving the first best: some special cases. PROPOSITION 1. *When the two policy authorities cooperate, the first best can be achieved following productivity, net worth, and incentive-constraint shocks.*

PROOF. Absent markup shocks, the solution that implies $\hat{\pi}_t = 0$, $\hat{y}_t = \frac{1+\theta}{\sigma+\theta}\hat{a}_t$, $\hat{R}_t = -\sigma\frac{1+\theta}{\sigma+\theta}(1-\rho_a)\hat{a}_t$, and $\hat{\phi}_t = \sigma(1-\rho_a)\frac{(1+\phi)(1+\theta)}{(\sigma+\theta)}\hat{a}_t$ is consistent with the equilibrium conditions (22)–(26) and fully eliminates economic distortions—that is, from the loss function (27), we see that $\mathcal{L} = 0$. $\square$

Following a net worth shock, the policy prescription is trivial. The macroprudential subsidy offsets movements in network, by following a rule of this type

$$\hat{S}_t = -\frac{\hat{n}_t^s}{\delta - 1}.$$

So, the macroprudential authority subsidizes banks whenever those have a negative shock to their net worth. As a result, there are no movements in credit spreads, inflation, or interest rates, and the output gap remains closed. Similarly, the macroprudential authority can fully offset the effect of the shock to the incentive compatibility constraint λ_t so that inflation, spreads, and the output gap remain constant without the need to move the nominal interest rate.

In the case of productivity shocks, one can think of interest rates moving to counteract movements in inflation—as lower productivity always warrants an increase in interest rates (i.e., $\hat{R}_t = -\sigma\frac{1+\theta}{\sigma+\theta}(1-\rho)\hat{a}_t$)—and the macroprudential instrument is used to make sure spreads offset the effect of monetary policy on the total cost of borrowing for banks—for example, a contractionary monetary policy should be accompanied by a compression in spreads. This is a case where it is optimal for monetary policy to be tight and yet macroprudential policy to be loose. Some commentators cited in our introduction have suggested that this is a sign of coordination failure. On the contrary, we show that such a constellation is part of an optimal coordinated plan. The reason is simple. The two tools address different distortions and these distortion do not always comove cyclically in a way that requires loose policies for both tools.

Overall, the cooperative solution absent markup shocks achieves the first best because the two distortions—the nominal rigidity and the credit distortion—can be offset with the two available instruments. In fact, following shocks to the financial sector (namely, $\hat{n}_t^s$ and $\hat{\lambda}_t$), the macroprudential tool alone can fully offset the effect of those, leaving no inefficient macro-economic fluctuations.

PROPOSITION 2. *In the case of markup shocks, optimal policy ensures efficiency if $\alpha = 1$.*

PROOF. When there are only markup shocks and $\alpha = 1$, the solution that implies $\hat{\pi}_t = 0$, $\hat{y}_t = 0$, $\hat{R}_t = 0$, and $\hat{\phi}_t = -(1+\phi)\hat{\varepsilon}_t$ is consistent with the equilibrium conditions (22)–(26) and fully eliminates economic distortions—that is, from the loss function (27), we see that $\mathcal{L} = 0$. $\square$

Markup shocks affect entrepreneurs' costs in a fashion similar to the credit spreads. When $\alpha = 1$, the optimal macroprudential instrument moves to generate a spread that fully offsets the markup shock. In particular, it sets

$$\widehat{S}_t = \frac{(1 + \phi)\delta}{(\delta - 1)} \epsilon_t^\pi.$$

The macroprudential authority subsidizes banks' funding costs, which in turn implies lower spreads. As a result, there are no cost-push fluctuations for firms and the interest rate facing consumers can be kept constant by monetary policy. Note that, in the traditional New Keynesian model without credit frictions and one monetary policy instrument (replicated in our model by setting $\alpha = 0$ and $\widehat{S}_t = 0$), markup shocks introduce policy trade-offs and the first best cannot be achieved. So the aforementioned result illustrates that if one takes this model and introduces credit friction, the presence of a credit spread allows a planner with one instrument—the macroprudential one—to replicate the first best. Although there are three sources of inefficiencies (i.e., monopolistic markup fluctuations, credit frictions, and nominal rigidity), one can achieve the first best with one policy instrument. This is because macroprudential policy can engineer a credit spread that fully offset the cost push effects coming from markup shock. But when there are different factors of production (i.e., $\alpha \neq 1$) that are affected in different ways by the credit friction, the strategy of moving subsidies to offset cost-push fluctuations would generate misallocations of resources between such factors. The misallocation occurs, because the effective interest rate paid to hire one type of labor moves with the shock.

4.3 Noncooperative Equilibrium

This section studies the equilibrium under discretion when policy is set in a noncooperative game between the two institutions. The case of noncooperative equilibria under commitment is discussed in Section 5.

Delegation of loss functions. As discussed, and documented in the monetary policy literature, the lack of a commitment technology—which would enable the policy-maker to achieve a desired course for expectations—worsen the policy trade-offs and generate welfare losses. The literature following Rogoff 1985 has discussed how delegating objective functions to policymakers that are different than the social welfare function can improve the equilibrium allocation under discretion. In this section, we consider different delegations of objectives for the central bank and the macroprudential authority. In particular, we consider the following objectives for the central bank ($\mathcal{L}_t^{cb}$) and the macroprudential authority ($\mathcal{L}_t^{mp}$).

$$\mathcal{L}_t^{cb} = E_t \sum_{i=0}^{\infty} \beta^i \left[\frac{1}{2} \widehat{\pi}_{t+i}^2 + \chi_y^{cb} (\widehat{y}_{t+i}^g)^2 \right], \quad (34)$$

$$\mathcal{L}_t^{mp} = E_t \sum_{i=0}^{\infty} \beta^i [\chi_y^{mp} (\hat{y}_{t+i}^g)^2 + (\hat{f}_{t+i})^2]. \quad (35)$$

While in the benchmark specification used in our analysis we set χ_y^{cb} and χ_y^{mp} so that the relative weights between the two objectives for each authority equals the ones in the social loss—that is, we set $\chi_y^{cb} = \chi_y$ and $\chi_y^{mp} = \chi_y / \chi_f$ —, we also consider the implications of setting different values for such weights. By considering the delegations described above in which each authority have one or two economic distortions in its loss function, we believe we are covering most relevant delegations. One can also argue that our benchmark delegation is consistent with arrangements currently being put in practice by central banks and regulatory authorities. For example, the objectives of the Bank of England's Financial Policy Committee are related to the resilience of the financial system as a primary goal while supporting economic policy as a secondary goal. The Bank of England's Monetary Policy Committee aim at ensuring price stability as a primary goal while also supporting economic policy as a secondary goal objective. Obviously, one could consider a loss function that depends on yet other variables, but these would be inherently *ad hoc* specifications.

In what follows, we study different policy arrangements. We first consider the case in which the authorities play a Nash game, that is, they simultaneously choose their policy instruments taking as given the reaction function of the other authority. We then study whether the coordination problem can be mitigated if one authority could lead the decision-making process within the period. In fact, given that macroprudential policy is often seen to move at a lower frequency than monetary policy, one crude way to capture such difference in frequency is to let the macroprudential instrument be chosen first within the period, that is, the macroprudential authority leads the monetary authority. We also investigate leadership by the monetary authority.

Nash equilibrium. We now characterize the Nash equilibrium in which each policy authority chooses its instruments taking as given the reaction function of the other authority.⁷ The problem of the monetary policy authority is to choose $\hat{\pi}_t$, $\hat{y}_t$, $\hat{f}_t$, $\hat{R}_t$, $\hat{n}_t$, $\hat{\delta}_t$, and $\hat{\phi}_t$ to maximize $\mathcal{L}_t^{cp}$, taking $\hat{S}_t$ as given, while the problem of the macroprudential authority is to choose $\hat{\pi}_t$, $\hat{y}_t$, $\hat{f}_t$, $\hat{S}_t$, $\hat{n}_t$, $\hat{\delta}_t$, and $\hat{\phi}_t$ to maximize $\mathcal{L}_t^{mp}$, taking $\hat{R}_t$ as given. Both optimization problems have equilibrium conditions (22)–(26) as constraints.

PROPOSITION 3. *Macroprudential policy is independent of the weight of the output gap in the loss function $\mathcal{L}_t^{mp}$.*

PROOF. It can easily be shown that the solution to the Nash equilibrium implies that future inflation and the future output gap are independent of endogenous states. In this case, the output gap will be fully determined by the central bank's choice of interest rates and terms independent of policy. Hence, the macroprudential authority considers the output gap to be a term independent of its policy. As a result, the macroprudential

7. We consider only equilibria in pure strategies. We thank an anonymous referee for pointing this out.

authority chooses to fully eliminate fluctuations in the effective interest rates (i.e., $\widehat{f}_t = 0$), regardless of the weight of the output gap in its loss function. $\square$

The problem of the monetary authority can be characterized as follows. We conjecture again an equilibrium where none of the welfare-relevant variables $\{\pi_t, y_t^g, f_t\}$ depend on the endogenous state variable. Hence, the continuation value function of the central bank is independent of current central bank policy. However, unlike the problem under cooperation, the Euler equation and the banking sector equations (23)–(26) now become binding constraints for the monetary authority. The problem is

$$\min \quad \widehat{\pi}_t^2 + \chi_y^{cb} (\widehat{y}_t^g)^2 + t.i.p.$$

Here, the expression *t.i.p.* denotes that the omitted terms are terms independent of policy. The minimization is subject to

$$\begin{aligned}\widehat{\pi}_t &= \kappa [(\sigma + \theta)\widehat{y}_t^g + \alpha\widehat{f}_t] + t.i.p., \\ \widehat{R}_t &= -\sigma\widehat{y}_t^g + t.i.p., \\ \widehat{n}_t^r &= -\widehat{\pi}_t + t.i.p., \\ \widehat{\delta}_t &= -\widehat{n}_t^r + (1 + \sigma + \theta)\widehat{y}_t^g - (1 - \alpha)\widehat{f}_t + t.i.p., \\ \widehat{\delta}_t &= \delta\widehat{\phi}_t + (\delta - 1)\widehat{S}_t - \beta(\phi\delta + 1)x_t + t.i.p., \\ \widehat{f}_t &= \widehat{R}_t + b\widehat{\phi}_t.\end{aligned}$$

In the last equation, we have substituted in our conjecture that future leverage δ_{t+1} depends on the endogenous state x_t defined by equation (30) with coefficient -1 . It can be verified that this must be true in any equilibrium where $\{\pi_t, y_t^g, f_t\}$ do not depend on x_{t-1} .

The solution to the problem can be expressed as a targeting rule

$$\widehat{\pi}_t = -\Theta\widehat{y}_t^g. \quad (36)$$

Here, the coefficient Θ is given by

$$\Theta = \left(\frac{\chi_y^{cb}}{\kappa} \right) \frac{1 - \alpha + (1 - \beta)(\delta - \alpha\kappa) + \phi(1 - \beta\alpha)\delta}{\mu + (1 - \beta + \phi)\delta\mu + (1 - \beta)\alpha(\sigma - \delta - \delta\phi)},$$

where $\mu \equiv \sigma + \theta + \alpha(1 - \sigma)$. We can obtain a simpler, but only approximate targeting criterion by looking at the limit where $\beta = 1$ and $\phi = 0$:

$$\Theta \sim \left(\frac{\chi_y^{cb}}{\kappa} \right) \frac{(1 - \alpha)}{\sigma + \theta + \alpha(1 - \sigma)}.$$

Relative to the targeting criterion under cooperation, in the Nash equilibrium the central bank acts more inflation averse. To see this, note that the coefficient in the targeting criterion in the Nash equilibrium is a fraction of the corresponding coefficient under cooperation, where that fraction is given by

$$0 < \frac{1}{1 + \frac{\alpha(1+\theta)}{(1-\alpha)(\sigma+\theta)}} < 1.$$

The reason the targeting criterion of the central bank in the Nash equilibrium is more inflation averse is as follows. Under cooperation, the impact of the working capital constraint on inflation dynamics (given by f_t in the Phillips curve) does not affect how the central bank trades off inflation and the output gap, because the optimal f_t can be chosen freely via the macroprudential instrument. In the Nash equilibrium, however, the central bank takes this instrument rule as given. It, therefore, has to take into account how the choice of the output gap affects inflation dynamics indirectly via f_t . In particular, the central bank perceives an increase in the output gap to raise inflation by $\kappa(\sigma + \theta)$ under cooperation but by $\kappa(\sigma + \theta) + \alpha(1 - \alpha)^{-1}(1 + \theta)$ in the Nash equilibrium.⁸ This, in turn, makes the central bank more inflation averse.

Although the Nash equilibrium implies a targeting criterion that puts higher weight on inflation stabilization than in the equilibrium under cooperation, as shown in Table 2, the welfare losses under the uncoordinated Nash equilibrium are almost twice as big as under cooperation. This is because of the full stabilization of spreads present in this environment. In fact, because the macroprudential authority does not attempt at stabilizing the output gap, it is not clear that the introduction of the macroprudential instrument is always advantageous. As shown in the last column of Table 2, the equilibrium in which there is no macroprudential instrument can be welfare superior to the Nash equilibrium. In particular, this is the case if we compare the setting in which the policymaker with one instrument (the nominal interest rates) puts a significant (and optimal) weight on stabilizing spreads with the case of a Nash game between two policy authorities whose loss functions have a small (and also optimal) weight of the output gap.

Clearly, one could consider introducing other variables in the macroprudential authority's objective. For example, we consider the case in which some "macroprudential gradualism" is introduced, that is, we assume that the loss function of the macroprudential authority increases with the volatility of its instrument:

$$\tilde{\mathcal{L}}_t^{mp} = E_t \sum_{i=0}^{\infty} \beta^i \left[(\hat{S}_{t+i})^2 + \chi_f^{mp} (\hat{f}_{t+i})^2 \right]. \quad (37)$$

We find that the introduction of the tool in the macroprudential loss can reduce welfare losses—as the policymaker moves away from fully stabilizing the effective

8. The additional term $\alpha(1 - \alpha)^{-1}(1 + \theta)$ is the monetary authorities perceived derivative of f_t with respect to the output gap. It is only the perceived derivative, because the macroprudential authority fully stabilizes f_t in equilibrium as suggested above.

interest rate—but the improvement is only marginal (using the optimal weights, the losses go from 0.0059% to 0.0058%).⁹

Monetary policy leadership. We now consider the case in which the central bank leads, and thus, takes into account how its policy choice affects the one of the macroprudential authority. Suppose that $\widehat{R}_t$ is set *within* the period before $\widehat{S}_t$. This is the so-called leadership equilibrium, which can be thought of as a within-period commitment by one policymaker. In this case, the instrument of the follower will depend on the instrument of the leader, as the latter is effectively predetermined within the period. The leader can thus choose a value on the followers' best response function and exploit this ability optimally.

PROPOSITION 4. *Under the benchmark assumption that the central bank's loss function has the same relative weight on inflation versus the output gap as the social planner—that is, $\chi_y^{cb} = \chi_y$, the central bank's policy rule under monetary policy leadership is identical to the one under cooperation, but the macroprudential authority chooses to fully stabilize fluctuations in the effective interest rate.*

PROOF. As in the Nash equilibrium, the solution under monetary policy leadership implies that inflation is independent of endogenous states (see the Technical Appendix available upon request), the output gap is fully determined by the central bank's choice of interest rates, and the best response of the macroprudential authority is to fully eliminate fluctuations in the effective interest rates (i.e., $\widehat{f}_t = 0$). The monetary policy authority takes this into account and solves the following problem

$$\min \quad \widehat{\pi}_t^2 + \chi_y^{cb} (\widehat{y}_t^g)^2 + t.i.p.$$

Subject to

$$\widehat{\pi}_t = \kappa(\sigma + \theta)\widehat{y}_t^g + t.i.p.,$$

$$\widehat{R}_t = -\sigma\widehat{y}_t^g + t.i.p.,$$

$$\widehat{n}_t^r = -\widehat{\pi}_t + t.i.p.,$$

$$\widehat{\delta}_t = -\widehat{n}_t^r + (1 + \sigma + \theta)\widehat{y}_t^g + t.i.p.,$$

$$\widehat{\delta}_t = \delta\widehat{\phi}_t - \beta(\phi\delta + 1)x_t + t.i.p.,$$

and the first-order condition combined can be represented by the following targeting rule:

$$\widehat{\pi}_t = -\frac{\chi_y^{cb}}{\kappa(\sigma + \theta)}\widehat{y}_t^g.$$

9. Clearly, one could consider introducing other variables and yet different specifications for macroprudential authority's objective. As previously discussed, we decided to restrict our attention to losses that depended in welfare relevant variables and that had some resemblance with the policy designed currently discussed by regulators and central banks.

This targeting rule is identical to the one under cooperation if $\chi_y^{cb} = \chi_y$. $\square$

As was demonstrated in the previous section, the targeting criterion under cooperation is less inflation averse than in the Nash equilibrium. Therefore, the same can be said about the Monetary Policy Leadership equilibrium. With a smaller weight on inflation stabilization (and the same target of fully stabilizing effective interest rates), the equilibrium under Monetary Policy Leadership leads to bigger welfare losses when compared with the Nash equilibrium as the inflation bias under discretion is more pronounced (see Table 2).

When the weight χ_y^{cb} is chosen optimally, it is identical to the one under cooperation (in particular, $\chi_y^{cb*} = 0.06$). Such weight can be thought of as the one that delivers the best trade-off between inflation and the output gap. Moreover, with optimal weights in place, welfare under Monetary Policy Leadership is identical to that under the Nash equilibrium. The two settings have the same general formulation for the optimal targeting rules (namely, they imply $f_t = 0$ and $\hat{\pi}_t = f(\cdot)\hat{y}_t^g$, where $f(\cdot)$ is a function of the weights in the loss function), so with optimized coefficients they ought to arrive at the same equilibrium. Note that the optimal weight in the Nash equilibrium is smaller than under Monetary Policy Leadership because, as previously discussed, in this equilibrium the central bank perceives the trade-off between inflation and output gap differently, and as a result, monetary policy is—for a given weight of inflation in the loss function—less tolerant to fluctuations in inflation.

Macroprudential leadership. In this section, we analyze the case in which the macroprudential authority leads. The targeting criterion of the central bank is identical to the one from the Nash game. As shown in the Technical Appendix available upon request, the problem of the macroprudential authority is

$$\min \quad \chi_y^{mp} (\hat{y}_{t+i}^g)^2 + (\hat{f}_{t+i})^2 + t.i.p.$$

subject to

$$\begin{aligned} \hat{\pi}_t &= \kappa [(\sigma + \theta)\hat{y}_t^g + \alpha \hat{f}_t] + t.i.p., \\ \hat{R}_t &= -\sigma \hat{y}_t^g + t.i.p., \\ \hat{n}_t^r &= -\hat{\pi}_t + t.i.p., \\ \hat{\delta}_t &= -\hat{n}_t^r + (1 + \sigma + \theta)\hat{y}_t^g - (1 - \alpha)\hat{f}_t + t.i.p., \\ \hat{\delta}_t &= \delta \hat{\phi}_t + (\delta - 1)\hat{S}_t - \beta(\phi\delta + 1)x_t + t.i.p., \\ \hat{f}_t &= \hat{R}_t + b\hat{\phi}_t, \end{aligned}$$

and

$$\hat{\pi}_t = -\Theta \hat{y}_t^g. \quad (38)$$

The last constraint is the best response of the central bank that the macroprudential authority—acting as a leader—takes into account when setting policy. The solution to the problem is

$$\widehat{y}_t^g = \frac{1}{\chi_y^{mp}} \left(\frac{(\sigma + \theta)}{\alpha} + \frac{\Theta}{\kappa\alpha} \right) \widehat{f}_t. \quad (39)$$

If $\chi_y^{mp} = \chi_y / \chi_f$

$$\widehat{f}_t = \frac{\chi_y}{\chi_f} \left(\frac{\alpha}{(\sigma + \theta) + \Theta/\kappa} \right) \widehat{y}_t^g. \quad (40)$$

The higher the weight on the output gap in the monetary policy authority loss function (higher χ_y^{cb} and thus higher Θ), the less the macroprudential authority will focus on output gap stabilization. As a result, the targeting rule (40) obtained under Macroprudential Leadership has a smaller weight on the output gap stabilization than the cooperative policy (33). Most important, as discussed earlier, the trade-off between inflation and the output gap dictated by the targeting rule (38) implies a higher weight on inflation stabilization than the one under cooperation. As a result, losses under cooperation are higher than under Macroprudential Leadership in our benchmark specification (see Table 2).

PROPOSITION 5. *If a social planner can freely choose optimally the weight in the loss function under cooperation (i.e., χ_y and χ_f) and the weights in the noncooperative setting (i.e., χ_y^{cb} and χ_y^{mp}), the Cooperative equilibrium and the Macroprudential Leadership equilibrium will be identical. This equilibrium is strictly superior, in terms of social welfare, to the solution under Monetary Policy leadership or the Nash equilibrium.*

PROOF. Equations (32) and (33) demonstrate that the cooperative solution can be written as

$$\begin{aligned} \widehat{\pi}_t &= f_c(\chi_y) \widehat{y}_t^g, \\ \widehat{f}_t &= g_c(\chi_y, \chi_f) \widehat{y}_t^g, \end{aligned}$$

and, similarly, equations (38) and (39) show that the solution under Macroprudential leadership can be written as

$$\begin{aligned} \widehat{\pi}_t &= f_{mpl}(\chi_y^{cb}) \widehat{y}_t^g, \\ \widehat{f}_t &= g_{mpl}(\chi_y^{mp}) \widehat{y}_t^g. \end{aligned}$$

If (χ_y, χ_f) or $(\chi_y^{cb}, \chi_y^{mp})$ are chosen as to minimize social losses (27), the solution for these two settings will be identical. Moreover, the equilibrium under the optimal weights will produce social losses no larger than in the case of Nash or Monetary

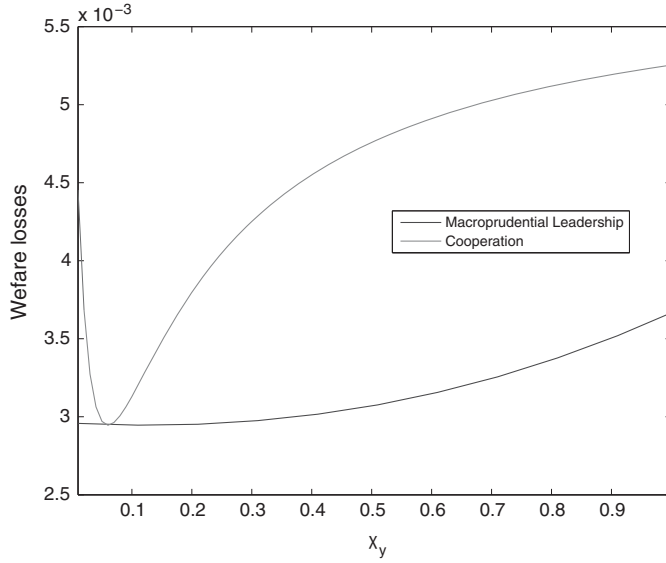


FIG. 1. Welfare as a Function of Relative Weight on the Output Gap.

Policy leadership, whose solution is restricted to the following targeting criteria:

$$\begin{aligned}\hat{\pi}_t &= f_{cbl, nash}(\chi_y^{cb})\hat{y}_t^g, \\ \hat{f}_t &= 0.\end{aligned}$$

□

Figure 1 shows how welfare losses under Macroprudential Leadership change with the weight of the output gap in the central bank's loss function changes. In this exercise, the weight of the spread in the macroprudential loss is kept at the level that minimizes welfare losses. The figures also shows how the losses under Cooperation change with the weight of the output gap in the loss function. Similarly, the weight of the effective interest rate in this loss function is kept at the optimal level.

The figure demonstrates that both optimal weights are low but, interestingly, it shows that cooperation does no better than Macroprudential Leadership under most parameter specifications. While with optimal weights, the loss is the same, when the the weight of the output gap in the central bank's loss is too low (high), Macroprudential Leadership does better because it compensates by putting a higher (lower) weight on output gap stabilization, see equation (39). Moreover, with optimal weights in place, the welfare losses with the macroprudential authority acting as a leader are virtually identical to the minimum level of losses under commitment.

5. SENSITIVITY

In this section, we perform sensitivity analysis to two natural variations. First, we analyze the possibility coordination problem in the case of commitment. Second, we introduce an alternative instrument for the macroprudential authority.

We also note that several qualitative results are not sensitive reasonable variations in the parameters governing financial frictions. That is, we have varied the steady-state spread and steady-state leverage around the baseline calibration and found that Macroprudential Leadership dominates the Nash equilibrium in welfare terms, which in turn dominates the equilibrium absent any macroprudential tool (with monetary policy set optimally to maximize social welfare).¹⁰ In addition, one may be interested in a setting where the monetary authority follows a simple Taylor rule and the macroprudential authority acts optimally under discretion. In response to productivity shocks, this results in very small welfare losses as the Taylor rule with standard coefficients does not fully stabilize inflation. In response to financial shocks, there are zero welfare losses because they can be fully offset by macroprudential policy alone. The welfare losses in response to markup shocks are of very similar magnitude as in the discretionary Nash equilibrium.

5.1 Comparison to Commitment

The solution under commitment serves as a natural point of comparison for the results under discretion.¹¹ However, we have found there typically does not exist a Nash equilibrium under commitment when the central bank chooses the nominal interest rate as its instrument. This finding is familiar from the open economy monetary policy literature; see Coenen et al. (2007). One can, however, compare discretion and commitment in a noncooperative setting, if the central bank chooses the money growth rate as its instrument, as an equilibrium typically exists in this case. We assume a demand function for real money balances $\hat{l}_t$ of the following form:

$$\hat{y} - \eta \hat{R}_t = \hat{l}_t, \quad (41)$$

where η is the interest elasticity of money demand, which we set to 4 in our numerical exercise, as in Gali (2008). This can be derived from a money-in-the-utility-function framework. If we assume that the coefficient of real money balances in the utility functions become arbitrarily small, our quadratic welfare criterion is unaffected. From

10. In particular, leverage and spreads are increased threefold relative to the baseline calibration and reduced to a third of that calibration. We have also varied the degree of price stickiness and found that our results are robust.

11. The typical equilibrium concept under commitment is open-loop, mainly because this equilibrium is easy to compute. In an open-loop equilibrium, both policy policymakers decide an entire *path* for their instruments at the beginning of time. This contrasts with policymakers choosing a reaction *function* under a closed-loop or feedback equilibrium.

TABLE 3
WELFARE LOSSES (MONEY GROWTH RATE INSTRUMENT)

	Cooperation	Nash
Commitment	0.0028	0.0030
Discretion	0.0044	0.0073

the definition of real money balance, one can write the following identity linking it to the growth rate of the nominal money stock Δm_t set by the central bank:

$$\Delta \widehat{l}_t = \Delta \widehat{m}_t - \widehat{\pi}_t. \tag{42}$$

Table 3 summarizes the welfare losses arising with this instrument for the case of discretion and commitment

The table reveals that noncooperative settings can result in larger gains from cooperation under discretion than under commitment. This is because the two frictions of limited commitment and uninternalized spillovers interact in an adverse way under discretion. This is in line with our result that, once we allow for a smaller weight of the output gap in the central bank’s loss function—which reduces the so-called discretionary bias—the gains from cooperation also fall. We should note that the change of monetary policy instrument, from interest rate to money growth, does not alter the ranking (or magnitude) of welfare losses shown in Table 2.

5.2 Alternative Policy Instrument

There is a lot of debate in policy circles about the appropriate instrument for counteracting financial sector distortions. In the baseline model, we have assumed that the instrument is a subsidy to the funding costs of banks. Here, we are considering an alternative instrument that shares some similarities with a liquidity requirement. We assume banks are required to invest a fraction G_t of their deposits into other riskless assets instead of using them to provide funding for firms. We abstract from modeling explicitly which activities these assets are funding instead, although we think about them as being invested in government bonds or as remunerated reserves. We assume that the bank earns the riskless interest rate on the assets and pays these earnings out to depositors, so there is no direct effect on banks’ profits. However, for a given size of the balance sheet and with a higher “liquidity requirement,” the bank earns its interest margin on a smaller volume of loans to the private sector and this reduces the franchise value. Hence, a higher liquidity requirement results in less loans to firms directly and indirectly. Conversely, a lowering requirement boots lending to firms, *ceteris paribus*. As in our baseline analysis, we assume that this steady-state effect is offset with a commensurately higher wage subsidy for households so that the steady state remains efficient.

TABLE 4
WELFARE LOSSES WITH ALTERNATIVE INSTRUMENT

	Commitment		Discretion		
	Coop.	Coop.	Nash	CB leads	MP leads
Welfare losses	0.0028	0.0044	0.0118	0.0113	0.0062

The net worth of a given banker j is given by

$$\begin{aligned} N_{jt} &= R_{t-1}^B B_{jt-1} + R_{t-1}^A G_{t-1} A_{jt-1} - R_{t-1}^A A_{jt-1} \\ &= R_{t-1}^B B_{jt-1} - R_{t-1}^A (1 - G_{t-1}) A_{jt-1}. \end{aligned} \quad (43)$$

We assume that the bank can run away with a fraction λ^B of the loans and a fraction λ^G of the deposits held as reserves. So equation (12) becomes

$$V_{jt} \geq \lambda^B B_t + \lambda^G G_t A_t. \quad (44)$$

One can show that the accumulation equation for bank capital and the incentive constraint in log-linear form can be expressed as follows while all other equations of the system (22)–(26) remain unchanged:

$$\hat{n}_t^r = \hat{n}_{t-1}^r + \hat{R}_{t-1} - \hat{\pi}_t + \frac{1}{\phi\delta + 1} [\delta\phi\hat{\delta}_{t-1} + \delta\hat{\phi}_{t-1}] + \hat{n}_t^s, \quad (45)$$

$$\hat{\delta}_t = (1 - gf_g)\delta\hat{\phi}_t - f_g\hat{g}_t + \gamma\beta\pi_b E_t[(1 + \delta\phi - gf_g\delta\phi)\hat{\delta}_{t+1} + f_g\hat{g}_{t+1}], \quad (46)$$

where leverage δ_t is now defined as loans to the *private* sector relative to bank capital.

It is straightforward to show that most results with a funding subsidy as the macroprudential tool will obtain in this setting as well. Differences between cooperative and noncooperative settings will arise only for markup shocks. Furthermore, under discretion with markup shocks the cooperative solution is identical as with the baseline tool. In both cases, the problem boils down to choosing f_t , y_t , π_t subject to the Phillips curve only. Furthermore, the case of monetary leadership is also identical. In this case, the problem of the monetary authority again boils down to choosing π_t , y_t subject to the Phillips curve and the reaction function of the macroprudential authority as the follower: $f_t = 0$. The only difference arises in the Nash equilibrium and with macroprudential leadership. The reason is simple. The targeting criterion of the monetary authority that takes the macroprudential reaction function as given is different. This difference arises because the incentive constraint now contains the *future* macroprudential instrument. The monetary authority takes as given how this future instrument of the financial regulator depends on the endogenous state, while no such consideration is present in the case of the funding subsidy as the tool. Table 4 summarizes the numerical results for the markup shock.

As can be seen, overall the results are very similar as with the baseline instrument.

6. CONCLUSION

This paper focuses on coordination issues that might arise when both monetary and macroprudential policy set instruments that affect credit conditions but they have different policy objectives. Whether policy is set in a coordinated fashion or not does not matter when the economy is only faced with productivity or financial shocks. In this case, the two instruments can fully eliminate economic inefficiencies, regardless of the monetary and macroprudential policy arrangement. Macroprudential policy is also found to be a powerful instrument when financial frictions equally affect the financing of different factors of production. In this case, the regulatory tool alone can eliminate inefficiencies coming from markup shocks. But when the first best cannot be achieved, we find that leadership by the macroprudential authority will lead to welfare outcomes that are as large as those achieved when the authorities cooperate—if not larger. This seems like an encouraging result, given that, in reality, macroprudential policy is set by a government body with a slower process of decision making. That is, it is reasonable to think that monetary policy decisions take regulators rules and decisions as given when formulating their own, but the reverse might not necessarily be the case. Another result with a direct practical relevance is the fact that, in our model, a macroprudential tool that affects funding costs or liquidity requirements tends to work in a similar fashion and arrive at similar outcomes.

Our analysis considers the interaction between policy instruments in a classic business cycle framework, but one can conceive of macroprudential policy as guarding against disaster risk that occurs very infrequently. Another interesting and relevant exercise would be to analyze whether our results would be robust to changes in the specification of inefficiencies in financial intermediations, such as the ones created by liquidity impairments.

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